THE COMPLEXITY OF RECOGNIZING MF GROUPS FROM FINITE PRESENTATIONS

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A countable group is MF if it admits finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. We show that the set of finite presentations of MF groups is Π^0_2-complete in the arithmetical hierarchy, and the set of finite presentations of non-MF groups is Σ^0_2-complete; in particular neither set is recursively enumerable. The upper bound is a description of the MF property by finite certificates. The lower bound encodes the set of programs with infinite domain into finite presentations by an effective Higman embedding, a central HNN extension, and a twisted one, using the non-MF group of [NMF] on one branch and, on the other, an MF permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona.

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1. Introduction

Not every countable group is MF: *Non-MF Groups* [NMF] constructs a finitely generated simple group $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ with property (T) such that every homomorphism from H to an MF group is trivial. We determine how hard the MF property is to recognize from a finite presentation, in the Kleene–Mostowski arithmetical hierarchy whose levels Σ^0_n and Π^0_n count alternations of quantifiers over a decidable relation.

Theorem A (MF recognition is Π^0_2-complete)**.** *Fix an effective coding of finite presentations of groups by natural numbers. The set of codes of finite presentations of MF groups is Π^0_2-complete, and the set of codes of finite presentations of non-MF groups is Σ^0_2-complete. In particular, neither set is recursively enumerable, and no algorithm decides from a finite presentation whether the presented group is MF. More precisely, there is a computable map $e \mapsto \widehat{R}_e$ into finite presentations such that the group presented by $\widehat{R}_e$ is MF if and only if the e-th partial computable function has infinite domain.*

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Recognizing the MF property is therefore exactly as hard as deciding whether a program halts on infinitely many inputs, and remains undecidable relative to an oracle for the halting problem. Figure 1 places the theorem in the arithmetical hierarchy.

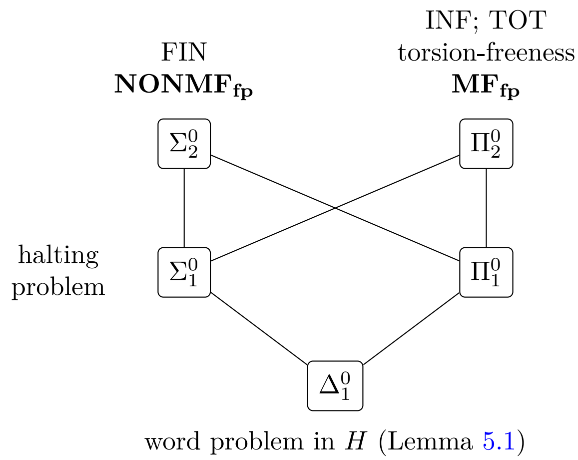

Figure 1: The arithmetical hierarchy. Lines are inclusions, and every listed problem is complete at its level; the word problem in H is decidable. Boldface marks the two sets of Theorem A: finite certificates put $\mathrm{MF}_{\mathrm{fp}}$ in Π^0_2 (Section 3), and the family $\widehat{R}_e$ reduces INF to $\mathrm{MF}_{\mathrm{fp}}$ and FIN to $\mathrm{NONMF}_{\mathrm{fp}}$ (Section 5).

Proposition 3.2 gives a Π^0_2 description of the MF presentations: a presentation is MF exactly when finite certificates, each checkable by decidable real arithmetic, exist at every scale (Section 3). For hardness we reduce the Π^0_2-complete set of programs with infinite domain to MF recognition: from e we construct a finite presentation $\widehat{R}_e$ (Section 5). If the domain is finite, then $H \leq \widehat{R}_e$, so $\widehat{R}_e$ is not MF. If the domain is infinite, then $\widehat{R}_e$ satisfies the hypotheses of Theorem 4.2, an MF permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona, and is MF (Sections 4 and 5). The MF property is closed under direct limits, as Korchagin proved [Kor, Proposition 13], so every finitely generated group that is not MF is a quotient of a finitely presented one; Theorem A gives the uniform family and the exact level.

Relation to prior work. Decision problems for finitely presented groups go back to Dehn, and the theorem of Adian and Rabin [Ad, Ra] shows that no Markov property (one that some finitely presented group has and that no group containing some fixed finitely presented group has) can be recognized from a finite presentation. Once a finitely presented non-MF group exists, the MF property is Markov, and Adian–Rabin gives undecidability; Theorem A gives the exact level. A classical Π^0_2-complete recognition problem for finitely presented groups is torsion-freeness, by Lempp [Lem]. For recursively presented groups, Bilanovic–Chubb–Roven proved that every Markov property is Π^0_2-hard to recognize [BCR]; their construction is the first step of Section 5. The passage from recursive to finite presentations is where the operator algebra enters: a generic Higman embedding does not preserve the MF property, and the HNN extensions of Section 5 are chosen so that Theorem 4.2 applies to them.

2. MF groups

A countable group G is *MF* if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE]. Here, MF group always means this operator-norm local-approximation property. The strong-convergence convention also requires the matrix models to recover the norms of the left regular representation [GKM, Sch]. Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}),$$

where $\bigoplus_n M_{d_n}(\mathbb{C})$ is the c_0-direct sum of the sequences with $\|x_n\| \to 0$. A separable C^*-algebra is called MF if it embeds as a C^*-subalgebra of a norm matrix corona [BK]. Restricting models to a subgroup shows that subgroups of MF groups are MF.

Throughout, the binary Leavitt algebra $L_{\mathbb{F}_2}(1,2)$ is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to $t_i s_j = \delta_{ij}$ and $s_0 t_0 + s_1 t_1 = 1$, and $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ is the subgroup of $\mathrm{GL}_{12}(L_{\mathbb{F}_2}(1,2))$ generated by the elementary matrices $e_{ij}(a) = 1 + aE_{ij}$, $i \neq j$. By [NMF, Theorem B], every homomorphism from H to an MF group is trivial; in particular H is not MF.

3. Locality and finite certificates

The MF property is defined through sequences of models on all of G, but it is determined by finite data with a fixed separation constant. The reformulation is due to Korchagin [Kor].

The following finite-domain formulation is derived from Korchagin [Kor, Proposition 7].

Lemma 3.1 (locality with fixed separation)**.** *A countable group G is MF if and only if for every finite set $F \subseteq G$ containing 1 and every $\varepsilon > 0$ there are $d \geq 1$ and a map $V\colon F \to \mathcal{U}(d)$ with $V(1) = 1$ such that*

$$\|V(gh) - V(g)V(h)\| \leq \varepsilon \quad \textit{whenever } g, h, gh \in F,$$

$$\|V(g) - 1\| \geq 1 \quad \textit{for } g \in F \setminus \{1\}.$$

Proof. Maps for an exhaustion $(F_n, 1/n)$ of G, extended by 1 outside F_n, form an MF model. Conversely, Korchagin's amplification — replacing $V_n(g)$ by a direct sum of its tensor powers — raises the separation of finitely many nonidentity elements above 1 while multiplying the defects by a constant [Kor, Proposition 7]. A block direct sum of late amplified models, one summand for each $g \in F \setminus \{1\}$, gives the stated map. □

We use the Kleene–Mostowski arithmetical hierarchy in its standard form [So]: a set $A \subseteq \mathbb{N}$ is Σ^0_1 if $A = \{x : \exists y\, R(x, y)\}$ for a decidable relation R, it is Π^0_1 if its complement is Σ^0_1, and Σ^0_2 and Π^0_2 are the classes $\{x : \exists y\, \forall z\, R(x, y, z)\}$ and $\{x : \forall y\, \exists z\, R(x, y, z)\}$ with R decidable. Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_P for the group presented by the code P. Let $\mathrm{MF}_{\mathrm{fp}}$ be the set of codes P such that G_P is MF, and $\mathrm{NONMF}_{\mathrm{fp}}$ its complement. Korchagin's Proposition 8 [Kor] is the finite-presentation form of Lemma 3.1; the certificates below add normal-closure witnesses for the trivial words, which is what makes the relation decidable.

Proposition 3.2 (finite certificates)**.** *There is a decidable relation $C(P, n, c)$ on triples of natural numbers such that a code P lies in* $\mathrm{MF}_{\mathrm{fp}}$ *if and only if for every n there is c with $C(P, n, c)$. Consequently* $\mathrm{MF}_{\mathrm{fp}} \in \Pi^0_2$ *and* $\mathrm{NONMF}_{\mathrm{fp}} \in \Sigma^0_2$.

Proof. Let $P = \langle x_1, \dots, x_k \mid r_1, \dots, r_m\rangle$. A *certificate at scale n* is a triple $c = (d, \ell, \pi)$ consisting of a dimension $d \geq 1$, a labelling ℓ of every reduced word of length at most n in $x_1^{\pm 1}, \dots, x_k^{\pm 1}$ by one of the symbols T and S, and, for every word w labelled T, an expression of w as a product of conjugates of the relators and their inverses that freely reduces to w. Such an expression is checkable, and its existence forces $w = 1$ in G_P. Let $\Phi(P, n, c)$ be the statement that there are $U_1, \dots, U_k \in \mathcal{U}(d)$ with $\|r_i(U) - 1\| \leq 2^{-n}$ for $i \leq m$ and $\|w(U) - 1\| \geq 1/4$ for every word w labelled S, where $w(U)$ is the evaluation of w at $U_1, \dots, U_k$ with $x_j^{-1} \mapsto U_j^*$. Writing $U_j = A_j + iB_j$ with real matrices, unitarity is a system of polynomial equations, and a bound $\|X\| \leq t$ or $\|X\| \geq t$ on a matrix whose entries are polynomials in the variables is a semialgebraic condition. Indeed, $\|X\| \leq t$ holds precisely when $t^2 - X^*X$ is positive semidefinite, while, for $t \geq 0$, $\|X\| \geq t$ holds precisely when there is a vector v with $\|v\| = 1$ and $\|Xv\|^2 \geq t^2$. Thus $\Phi(P, n, c)$ is a sentence of the first-order theory of the ordered field of real numbers, and its truth is decidable by Tarski's theorem [Ta]. Let $C(P, n, c)$ hold when c is a well-formed certificate at scale n whose T-expressions check and $\Phi(P, n, c)$ holds. Then C is decidable.

Suppose that G_P is MF. Fix n, let $L = \max(\{1, n\} \cup \{|r_i| : 1 \leq i \leq m\})$, let F be the set of images in G_P of all words of length at most L, together with their inverses, and let $\varepsilon = 2^{-n}/(4L)$. Lemma 3.1 gives $V\colon F \to \mathcal{U}(d)$ with $V(1) = 1$, defect at most ε on F, and $\|V(g) - 1\| \geq 1$ for $g \in F \setminus \{1\}$. Put $U_j = V(\bar{x}_j)$, where $\bar{w}$ denotes the image of a word w in G_P. For $g \in F$ we have $g^{-1} \in F$ and $\|V(g)V(g^{-1}) - 1\| \leq \varepsilon$, so $\|V(g^{-1}) - V(g)^*\| \leq \varepsilon$. For a word $w = y_1 \cdots y_\ell$ of length $\ell \leq L$, all of whose prefixes have images in F, induction on ℓ gives $\|w(U) - V(\bar{w})\| \leq 2\ell\varepsilon$. In particular $\|r_i(U) - 1\| \leq 2L\varepsilon \leq 2^{-n}$. Label a word w of length at most n by T if $\bar{w} = 1$, with a normal-closure expression as witness, and by S otherwise; for the latter, $\|w(U) - 1\| \geq \|V(\bar{w}) - 1\| - 2n\varepsilon \geq 1 - \frac{1}{2} \geq \frac{1}{4}$. So $C(P, n, c)$ holds for this certificate.

Conversely, suppose that for every n some certificate $c_n = (d_n, \ell_n, \pi_n)$ satisfies $C(P, n, c_n)$, and choose $U^{(n)} \in \mathcal{U}(d_n)^k$ witnessing $\Phi(P, n, c_n)$. For each $g \in G_P$ fix a reduced word w_g representing g, with w_1 the empty word, and put $V_n(g) = w_g(U^{(n)})$. Then $V_n(1) = 1$. For $g, h \in G_P$ the word $t = w_g w_h w_{gh}^{-1}$ is trivial in G_P, so it is freely equal to a product of A conjugates of relators and their inverses, for some A depending on g and h but not on n. Since evaluation is a homomorphism on the free group, since the operator norm is unitarily invariant, and since $\|xy - 1\| \leq \|x - 1\| + \|y - 1\|$ for unitaries x, y,

$$\|V_n(g)V_n(h) - V_n(gh)\| = \left\|t(U^{(n)}) - 1\right\| \leq A\,2^{-n} \longrightarrow 0.$$

For $g \neq 1$ the word w_g is nontrivial in G_P, so for $n \geq |w_g|$ the certificate c_n cannot label it T, hence labels it S, and $\|V_n(g) - 1\| \geq 1/4$. Thus $\limsup_n \|V_n(g) - 1\| \geq 1/4 > 0$, and G_P is MF.

The displayed equivalence exhibits $\mathrm{MF}_{\mathrm{fp}}$ as $\{P : \forall n\, \exists c\, C(P, n, c)\}$, a Π^0_2 set, and its complement as a Σ^0_2 set. ∎

4. HNN extensions with a corona conjugator

In [NMF] we prove non-approximability using one-sided conjugation in a norm matrix corona. If the edge isomorphism of an HNN extension is implemented by a unitary of a norm matrix corona, then the HNN extension is MF. For $\theta = \mathrm{id}$ this is close to Shulman's theorem on central HNN extensions [Sh]; the version below assumes a tracial state vanishing off the identity and allows a nontrivial edge isomorphism. The corona unitary implements the edge isomorphism, while the trace and the left regular representation prove that the group embeds in the resulting algebra.

A countable group G *admits a tracial MF realization* (A, ρ, τ) if A is a separable unital MF C^*-algebra, $\rho\colon G \to \mathcal{U}(A)$ is a homomorphism, and τ is a tracial state on A with $\tau(\rho(g)) = 0$ for every $g \neq 1$. Since $\tau(1) = 1$, such a group embeds in $\mathcal{U}(A)$, so it is MF. Only the vanishing of τ off the identity is used below; traciality is what the constructions below provide. To embed an arbitrary norm matrix corona $\mathcal{Q}_{\mathbf{d}}$ in $\prod_k M_k(\mathbb{C}) / \bigoplus_k M_k(\mathbb{C})$, choose a strictly increasing sequence (m_n) with $m_n \geq d_n$, embed $M_{d_n}(\mathbb{C})$ as a corner of $M_{m_n}(\mathbb{C})$, and put zero in the unused coordinates. The resulting map on reduced products is an isometric $*$-homomorphism, so the MF algebras of Section 2 are exactly the separable C^*-algebras that embed in the latter corona, as in [BK, Sh].

Lemma 4.1 (residually finite groups)**.** *Every countable residually finite group admits a tracial MF realization.*

Proof. Choose finite-index normal subgroups $N_1 \geq N_2 \geq \cdots$ with trivial intersection, and let λ_k be the left regular representation of G/N_k, of dimension d_k. In the corona $\mathcal{Q}_{\mathbf{d}}$ the map $\rho(g) = [(\lambda_k(g))_k]$ is a homomorphism, and $\tau([(x_k)_k]) = \lim_\omega \operatorname{tr}_{d_k}(x_k)$, for a free ultrafilter ω, is a tracial state with $\tau(\rho(g)) = 0$ for $g \neq 1$, since $g \notin N_k$ for all large k. The C^*-algebra generated by $\rho(G)$ and the unit of the corona is separable, unital, and MF, and carries the required realization. $\square$

Let G be a countable group, let $S \leq G$ be a subgroup, and let $\theta\colon S \to G$ be an injective homomorphism. The HNN extension

$$R = \langle G, t \mid tst^{-1} = \theta(s)\ (s \in S)\rangle \tag{1}$$

contains G by Britton's lemma [Bri].

Theorem 4.2 (HNN permanence with a corona conjugator)**.** *In the situation of (1), suppose that G admits a tracial MF realization (A, ρ, τ), and that for some norm matrix corona $\mathcal{Q}$ there are an injective $*$-homomorphism $\iota\colon A \to \mathcal{Q}$ and a unitary $W \in \mathcal{Q}$ with*

$$W\,\iota\rho(s)\,W^* = \iota\rho(\theta(s)) \qquad (s \in S).$$

Then R admits a tracial MF realization. In particular, R is MF.

Proof. Write $D = C^*(\iota\rho(G)) \subseteq \mathcal{Q}$, a separable C^*-algebra with unit $p = \iota(1)$, and let $B_0 = C^*(\iota\rho(S))$ and $B_1 = C^*(\iota\rho(\theta(S)))$, both containing p. Conjugation by W restricts to a $*$-isomorphism $\Theta\colon B_0 \to B_1$ carrying $\iota\rho(s)$ to $\iota\rho(\theta(s))$. Let U be the quotient of the full unital free product $D * C(\mathbb{T})$ by the closed ideal generated by the elements $ubu^* - \Theta(b)$, $b \in B_0$, where u is the canonical unitary generator of $C(\mathbb{T})$. We write d and u for the images of $d \in D$ and u in U. By construction, U has the following universal property: whenever $\pi\colon D \to E$ is a unital $*$-homomorphism into a unital C^*-algebra and $v \in \mathcal{U}(E)$ satisfies $v\pi(b)v^* = \pi(\Theta(b))$ for $b \in B_0$, there is a unique $*$-homomorphism $U \to E$ extending π and sending u to v. We first embed U in an amalgamated free product, which Shulman's criterion shows to be MF, and then construct the required tracial state on the copy of R inside U.

Step 1: U embeds in an amalgamated free product. This step follows a corner argument of Ueda [Ue]. Let $C = B_0 \oplus B_1$, $A_1 = M_2(D)$, and $A_2 = M_2(B_0)$, with the unital inclusions

$$\iota_A(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} \in A_1, \qquad \iota_B(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & \Theta^{-1}(c_1) \end{pmatrix} \in A_2,$$

and let $P = A_1 *_C A_2$ be the full amalgamated free product, in which A_1 and A_2 embed [Sh, Section 2.1]. Write e_{ij} and f_{ij} for the matrix units of A_1 and

A_2. Since $\iota_A(p,0)=\iota_B(p,0)$ and $\iota_A(0,p)=\iota_B(0,p)$, we have $e_{11}=f_{11}=: e$ and $e_{22}=f_{22}$ in P. Define $\pi_D\colon D\to ePe$ by $\pi_D(d)=\operatorname{diag}(d,0)$, a unital $*$-homomorphism into the corner, and put $w=e_{12}f_{21}\in ePe$. Then $ww^*=e_{12}f_{22}e_{21}=e_{12}e_{22}e_{21}=e$ and $w^*w=f_{12}e_{22}f_{21}=f_{12}f_{22}f_{21}=e$, so w is a unitary of ePe. For $b\in B_0$ the element $\operatorname{diag}(b,0)$ is $\iota_A(b,0)=\iota_B(b,0)$, so

$$\begin{aligned} w\pi_D(b)w^* &= e_{12}(f_{21}\operatorname{diag}(b,0)f_{12})e_{21} = e_{12}\,\iota_B(0,\Theta(b))\,e_{21}\\ &= e_{12}\,\iota_A(0,\Theta(b))\,e_{21} = \pi_D(\Theta(b)), \end{aligned}$$

where the middle equality uses $\iota_B(0,\Theta(b))=\operatorname{diag}(0,b)$ in A_2. The universal property of U gives a $*$-homomorphism $\Phi\colon U\to ePe\subseteq P$ with $\Phi|_D=\pi_D$ and $\Phi(u)=w$. We prove that Φ is injective by constructing $\Psi\colon P\to M_2(U)$ with $\Psi\Phi(x)=\operatorname{diag}(x,0)$. Let $\Psi_1\colon A_1\to M_2(U)$ be the entrywise inclusion $D\subseteq U$, and let $\Psi_2\colon A_2\to M_2(U)$ be the entrywise inclusion $B_0\subseteq U$ followed by conjugation by $\operatorname{diag}(1,u)$. On C we have $\Psi_1(\iota_A(c_0,c_1))=\operatorname{diag}(c_0,c_1)$ and $\Psi_2(\iota_B(c_0,c_1))=\operatorname{diag}(c_0,u\Theta^{-1}(c_1)u^*)=\operatorname{diag}(c_0,c_1)$, so the pair induces a $*$-homomorphism $\Psi\colon P\to M_2(U)$. On generators, $\Psi\Phi(d)=\operatorname{diag}(d,0)$ and $\Psi\Phi(u)=\Psi_1(e_{12})\Psi_2(f_{21})=\operatorname{diag}(u,0)$. Hence $\Psi\Phi$ is the injective $*$-homomorphism $x\mapsto\operatorname{diag}(x,0)$ of U into $M_2(U)$, and Φ is injective.

Step 2: P is MF. By Shulman's amalgamated-free-product criterion [Sh, Theorem 16], the full amalgamated free product $A_1 *_C A_2$ of separable C^*-algebras is MF as soon as there are injective $*$-homomorphisms φ_A and φ_B of A_1 and A_2 into one norm matrix corona with $\varphi_A\circ\iota_A=\varphi_B\circ\iota_B$. The first reduction inside this criterion is explicit: for a positive, strictly increasing matrix-size sequence, the closed C^*-subalgebra of the common corona generated by the two images is separable and MF, and the factor maps corestrict to compatible injective homomorphisms into it. Identify $M_2(\mathcal{Q})$ with the norm matrix corona of the doubled dimension sequence, and let $\varphi_A\colon M_2(D)\to M_2(\mathcal{Q})$ be the entrywise inclusion and $\varphi_B\colon M_2(B_0)\to M_2(\mathcal{Q})$ the entrywise inclusion followed by conjugation by $\operatorname{diag}(1,W)$. Both are injective, and on C,

$$\varphi_B(\iota_B(c_0,c_1)) = \begin{pmatrix} c_0 & 0\\ 0 & W\Theta^{-1}(c_1)W^*\end{pmatrix} = \begin{pmatrix} c_0 & 0\\ 0 & c_1\end{pmatrix} = \varphi_A(\iota_A(c_0,c_1)),$$

because Θ is conjugation by W. Hence P is MF, and so are its C^*-subalgebras ePe and $\Phi(U)\cong U$.

Step 3: the tracial state. The functional $T=\tau\circ\iota^{-1}$ is a tracial state on $\iota(A)\supseteq D$ with $T(\iota\rho(g))=0$ for $g\neq 1$. In the GNS representation of D for T, the vectors $\xi_g=\iota\rho(g)\xi_T$, $g\in G$, are orthonormal, because $\langle\xi_g,\xi_h\rangle=T(\iota\rho(h^{-1}g))$. Their closed span is invariant under D, and on it $\iota\rho(g)$ acts as the left regular representation $\lambda_G(g)$ of G. Thus there is a $*$-homomorphism $\pi\colon D\to C^*_{\mathrm{r}}(G)$ with $\pi(\iota\rho(g))=\lambda_G(g)$. Since $G\le R$, the space $\ell^2(R)$ is a direct sum of copies of $\ell^2(G)$ indexed by the right cosets of G, so $\lambda_R|_G$ is a multiple of λ_G and the assignment $\lambda_G(g)\mapsto\lambda_R(g)$ extends to a $*$-homomorphism $C^*_{\mathrm{r}}(G)\to C^*_{\mathrm{r}}(R)$. Composing, we obtain $\sigma_0\colon D\to C^*_{\mathrm{r}}(R)$ with $\sigma_0(\iota\rho(g))=\lambda_R(g)$. The pair $(\sigma_0,\lambda_R(t))$ satisfies the covariance relation

$\lambda_R(t)\sigma_0(b)\lambda_R(t)^* = \sigma_0(\Theta(b))$ for $b \in B_0$: on the generators $\iota\rho(s)$ both sides equal $\lambda_R(\theta(s))$, because $tst^{-1} = \theta(s)$ in R, and two $*$-homomorphisms that agree on generators agree on B_0. The universal property of U gives $\sigma\colon U \to C^*_{\mathrm{r}}(R)$ with $\sigma(\iota\rho(g)) = \lambda_R(g)$ and $\sigma(u) = \lambda_R(t)$.

Define $j\colon R \to \mathcal{U}(U)$ by $j(g) = \iota\rho(g)$ for $g \in G$ and $j(t) = u$. The defining relations of (1) hold in U, since $u\,\iota\rho(s)\,u^* = \Theta(\iota\rho(s)) = \iota\rho(\theta(s))$, so j is a homomorphism, and $\sigma \circ j = \lambda_R$ is injective, so j is injective. Let $A' = C^*(j(R)) \subseteq U$, a separable unital MF C^*-algebra, and let $\tau' = \tau_R \circ \sigma|_{A'}$, where τ_R is the canonical trace $x \mapsto \langle x\delta_1, \delta_1\rangle$ of $C^*_{\mathrm{r}}(R)$. Then τ' is a tracial state on A' with $\tau'(j(r)) = \tau_R(\lambda_R(r)) = 0$ for $r \neq 1$. Hence (A', j, τ') is a tracial MF realization of R. $\square$

Injectivity of j follows from $\sigma \circ j = \lambda_R$, not from the covariant pair $(\iota\rho, W)$, which need not be faithful; in the applications below it is not.

Corollary 4.3 (central HNN extensions)**.** *If G admits a tracial MF realization and $S \leq G$ is any subgroup, then $\langle G, t \mid [t, s] = 1\ (s \in S)\rangle$ admits a tracial MF realization.*

Proof. Apply Theorem 4.2 with θ the inclusion of S, with ι any injective $*$-homomorphism of A into a norm matrix corona, and with $W = 1$. $\square$

The positive branch of the reduction below produces tracial MF realizations inside norm reduced products. For C^*-algebras B_n, $n \geq 1$, write $\prod_n B_n / \bigoplus_n B_n$ for the quotient of the bounded sequences by the sequences with $\|x_n\| \to 0$; its norm is $\limsup_n \|x_n\|$.

Lemma 4.4 (reduced products of MF algebras)**.** *If every B_n is MF, then every separable C^*-subalgebra of $\prod_n B_n / \bigoplus_n B_n$ is MF. Moreover, $M_k(B)$ is MF whenever B is.*

Proof. The second assertion holds because M_k of a norm matrix corona is the norm matrix corona of the dimension sequence multiplied by k. For the first, let A be a separable C^*-subalgebra, and choose a sequence $(a_i)_{i\geq 1}$ dense in A, with lifts $(a_i^{(n)})_n$. Choose for each n an injective $*$-homomorphism η_n of B_n into a norm matrix corona, and lifts $(a_i^{(n,k)})_k$ of $\eta_n(a_i^{(n)})$ with $\sup_k \left\|a_i^{(n,k)}\right\| \leq \left\|a_i^{(n)}\right\| + 1$. Enumerate the noncommutative $*$-polynomials with coefficients in $\mathbb{Q}(i)$ as $p_1, p_2, \ldots$, and let $\mathcal{P}_n$ be the set of those among $p_1, \ldots, p_n$ that involve only $a_1, \ldots, a_n$, choosing $p_1 = 0$ so that $\mathcal{P}_n$ is nonempty. For $p \in \mathcal{P}_n$ the number $\left\|p(a^{(n)})\right\|$ equals $\limsup_k \left\|p(a^{(n,k)})\right\|$, since η_n is isometric. Hence there is K_n such that $\left\|p(a^{(n,k)})\right\| \leq \left\|p(a^{(n)})\right\| + 1/n$ for all $k \geq K_n$ and $p \in \mathcal{P}_n$, and for each $p \in \mathcal{P}_n$ there is $k_p \geq K_n$ with $\left\|p(a^{(n,k_p)})\right\| \geq \left\|p(a^{(n)})\right\| - 1/n$. Put

$$b_i^{(n)} = \bigoplus_{p \in \mathcal{P}_n} a_i^{(n,k_p)},$$

a block diagonal matrix. For $p \in \mathcal{P}_n$ the norm of $p(b^{(n)})$ is the maximum of $\left\|p(a^{(n,k_{p'})})\right\|$ over $p' \in \mathcal{P}_n$, which lies between $\left\|p(a^{(n)})\right\| - 1/n$ and $\left\|p(a^{(n)})\right\| + 1/n$. Every polynomial belongs to $\mathcal{P}_n$ for all large n, so $\limsup_n \left\|p(b^{(n)})\right\| = \limsup_n \left\|p(a^{(n)})\right\| = \|p(a)\|$ for every p. Therefore $a_i \mapsto [(b_i^{(n)})_n]$ is well defined and isometric on the $*$-algebra over $\mathbb{Q}(i)$ generated by the a_i, which is dense in A, and it extends to an isometric $*$-homomorphism of A into a norm matrix corona. $\square$

Lemma 4.5 (tensor synchronization)**.** *Let Γ be a countable group with a tracial MF realization (A_1, ρ_1, τ_1), let Q be a countable group with homomorphisms $\beta_n \colon Q \to B_n$ to finite groups such that every $q \neq 1$ satisfies $\beta_n(q) \neq 1$ for all large n, let $S \leq \Gamma$, and let $\tau \colon S \to Q$ be a homomorphism. Suppose that for every n there is a homomorphism $\lambda_n \colon \Gamma \to G_n$ to a finite group with*

$$\ker(\lambda_n|_S) \leq \ker(\beta_n \circ \tau).$$

Then there are a separable unital MF C^*-algebra A, a homomorphism $V \colon \Gamma \times Q \to \mathcal{U}(A)$, a tracial state T on A with $T(V(g,q)) = 0$ for $(g,q) \neq (1,1)$, and a unitary $W \in A$ with $WV(s,1)W^* = V(s,\tau(s))$ for all $s \in S$. Consequently

$$\langle \Gamma \times Q,\ u \ \mid\ u(s,1)u^{-1} = (s,\tau(s))\ (s \in S)\rangle$$

admits a tracial MF realization, and in particular is MF.

Proof. Let E_n be the image of $(\lambda_n, \beta_n) \colon \Gamma \times Q \to G_n \times B_n$, a finite group of order k_n, and let L_n be its left regular representation on $\ell^2(E_n)$. Put $B'_n = A_1 \otimes M_{k_n}(\mathbb{C})$, which is MF by Lemma 4.4, let $E = \prod_n B'_n / \bigoplus_n B'_n$, and define

$$V(g,q) = \left[\left(\rho_1(g) \otimes L_n(\lambda_n(g), \beta_n(q))\right)_n\right], \qquad T([(x_n)_n]) = \lim_\omega (\tau_1 \otimes \mathrm{tr}_{k_n})(x_n),$$

for a free ultrafilter ω. Each coordinate of V is a homomorphism into a unitary group, so V is a homomorphism, and T is a tracial state on E because $|(\tau_1 \otimes \mathrm{tr}_{k_n})(x_n)| \leq \|x_n\|$. If $g \neq 1$ then $\tau_1(\rho_1(g)) = 0$, so $T(V(g,q)) = 0$; if $g = 1$ and $q \neq 1$ then $\beta_n(q) \neq 1$ for all large n, the regular representation of a finite group has trace zero off the identity, and again $T(V(1,q)) = 0$.

The two homomorphisms $s \mapsto (\lambda_n(s), 1)$ and $s \mapsto (\lambda_n(s), \beta_n\tau(s))$ from S to E_n have the same kernel, namely $\ker(\lambda_n|_S)$, by the hypothesis. Their images are therefore isomorphic subgroups of E_n of the same order, isomorphic via $(\lambda_n(s), 1) \mapsto (\lambda_n(s), \beta_n\tau(s))$. The restriction of L_n to a subgroup of E_n is unitarily equivalent to the direct sum of as many copies of the left regular representation of that subgroup as it has right cosets, so there is a unitary $W_n \in M_{k_n}(\mathbb{C})$ with $W_n L_n(\lambda_n(s), 1) W_n^* = L_n(\lambda_n(s), \beta_n\tau(s))$ for all $s \in S$. Put $W = [(1 \otimes W_n)_n] \in E$; then $WV(s,1)W^* = V(s,\tau(s))$ for $s \in S$. Let A be the C^*-algebra generated by $V(\Gamma \times Q)$ and W; it is separable and unital, and MF by Lemma 4.4. The triple $(A, V, T|_A)$ is a tracial MF realization of $\Gamma \times Q$. For the last assertion, apply Theorem 4.2 to this realization. Let ι be an injective $*$-homomorphism of A into a norm matrix corona and put $p = \iota(1)$. The element $\widetilde{W} = \iota(W) + (1-p)$ is a unitary of the full corona and implements the required covariance on $\iota(A)$, so it is the required conjugator. $\square$

5. The complexity of recognizing MF groups

Write W_e for the domain of the e-th partial computable function and

$$\mathrm{INF} = \{e : W_e \text{ is infinite}\}, \qquad \mathrm{FIN} = \{e : W_e \text{ is finite}\}.$$

The set INF is Π^0_2-complete and FIN is Σ^0_2-complete [So, Chapter IV]. The construction combines the non-MF group H of [NMF] with Theorem 4.2: it produces a computable map $e \mapsto \widehat{R}_e$ into finite presentations such that the presented group contains H when $e \in \mathrm{FIN}$ and satisfies the hypotheses of Theorem 4.2 when $e \in \mathrm{INF}$, so that it is MF if and only if $e \in \mathrm{INF}$. For $e \in \mathrm{FIN}$, the group $\widehat{R}_e$ is not MF; for $e \in \mathrm{INF}$, Lemma 5.8 proves that it is MF. Table 1 lists the objects of the construction.

H	the non-MF group $\mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ of [NMF]
C_e	recursively presented; trivial for $e \in \mathrm{INF}$, contains H for $e \in \mathrm{FIN}$
$Q_e = F/N_e$	three-generated recursively presented group containing C_e; here $F = F(x,y,t)$
$Q_+ = F/N_+$	the value of Q_e for trivial C_e; embeds in $F(x_1,y) \times F(x_2,t)$
Λ_e; M_e	finitely presented Higman host of Q_e; the Mihailova subgroup detecting equality in Λ_e
K_e; L_e	a product of six free groups; its finitely generated subgroup with $i(F) \cap L_e = i(N_e)$
Γ_e; S_e	the central HNN extension $\langle K_e, v \mid [v,\ell] = 1\ (\ell \in L_e)\rangle$; $S_e = \langle i(F), v\, i(F)\, v^{-1}\rangle \cong F *_{N_e} F$
$\alpha_e \colon S_e \to Q_e$	q_e on $i(F)$, trivial on $v\, i(F)\, v^{-1}$
$\widehat{R}_e$	the finite presentation of $\langle \Gamma_e \times Q_e,\ u \mid u(s,1)u^{-1} = (s, \alpha_e(s))\ (s \in S_e)\rangle$

Table 1: The objects of the construction, in order of appearance.

A recursive presentation of H.

Lemma 5.1 (a recursive presentation of H). *The group $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ is finitely generated and has decidable word problem. In particular, it has a recursive presentation on finitely many generators.*

Proof. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. Since $e_{ij}(a+b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ for distinct i,j,k, induction on monomial length shows that $H = \langle Y\rangle$. The monomials in s_0, s_1, t_0, t_1 that contain no subword $t_i s_j$ and no subword $s_1 t_1$ form the standard reduced $\mathbb{F}_2$-basis of $L_{\mathbb{F}_2}(1,2)$ [Lev]. Indeed, with the orientation $t_is_j \to \delta_{ij}$ and $s_1t_1 \to 1 + s_0t_0$, the two critical words $t_is_1t_1$ and $s_1t_1s_j$ resolve to a common value, and the Diamond Lemma [Ber] gives these monomials as a basis. Products of reduced monomials are computed by the defining reductions, so equality in $L_{\mathbb{F}_2}(1,2)$ is decidable. A word in $Y^{\pm 1}$ is therefore evaluated to an explicit 12×12 matrix over $L_{\mathbb{F}_2}(1,2)$, and it represents 1 in H exactly when that matrix is the identity. The set of words representing 1 is decidable, hence recursively enumerable, and $\langle Y \mid \text{that set}\rangle$ is a recursive presentation of H. ☐

The group C_e.

Lemma 5.2 (the group C_e)**.** *There is an algorithm that computes from e a recursive presentation, on a computable set of generators, of a group C_e such that C_e is trivial if $e \in \mathrm{INF}$ and H embeds in C_e if $e \in \mathrm{FIN}$.*

Proof. Apply the construction of Bilanovic–Chubb–Roven [BCR, Theorem 3.1] to the trivial group and H. Let $\langle Y \mid T\rangle$ be the recursive presentation of H from Lemma 5.1. Take generators $y_{i,\ell}$ for $i \geq 1$ and $\ell \in Y$, one copy Y_i of Y for each i, and as relators all words of T rewritten in each copy Y_i, together with every generator $y_{i,\ell}$ for which $i \leq |W_e|$. The last family is recursively enumerable uniformly in e: enumerate W_e, and whenever its i-th element appears, enumerate all of Y_i. The group presented is the free product of the copies $H_i \cong H$, $i \geq 1$, with the first $|W_e|$ copies killed. If W_e is infinite every copy is killed and C_e is trivial; if W_e is finite, C_e is the free product of the remaining copies and contains H. ☐

A three-generator embedding. Let $F = F(x, y, t)$ be the free group on x, y, t, and let $y_i = x^iyx^{-i}$ for $i \in \mathbb{Z}$. For a countable group C with a generating sequence $(c_i)_{i\geq 1}$, put $c_i = 1$ for $i \leq 0$ and

$$B(C) = \langle C * F(x, y),\ t \ \mid\ ty_it^{-1} = c_iy_i\ (i \in \mathbb{Z})\rangle. \tag{2}$$

Let $Q_+ = B(1) = \langle x, y, t \mid [t, y_i] = 1\ (i \in \mathbb{Z})\rangle$, let $q_+\colon F \to Q_+$ be the quotient map, and let $N_+ = \ker q_+$.

Lemma 5.3 (the three-generator embedding)**.**

(1) *$B(C)$ is an HNN extension of $C * F(x, y)$, so C embeds in $B(C)$; it is generated by x, y, t; and a recursive presentation of $B(C)$ on x, y, t is computable from a recursive presentation of C. We write $B(C) = F/N_C$ and $q_C\colon F \to B(C)$ for the quotient map.*

(2) *Killing C induces a surjection $B(C) \to Q_+$; hence $N_C \leq N_+$, with equality when C is trivial.*

(3) *The assignment $x \mapsto (x_1, x_2)$, $y \mapsto (y, 1)$, $t \mapsto (1, t)$ defines an injective homomorphism $j\colon Q_+ \to P = F(x_1, y) \times F(x_2, t)$. In particular, Q_+ is residually finite.*

Proof. (1) The elements y_i, $i \in \mathbb{Z}$, freely generate the normal closure of y in $F(x, y)$ [LS, Chapter I]. The retraction $C * F(x, y) \to F(x, y)$ killing C sends c_iy_i to y_i, so a nontrivial reduced word in the c_iy_i maps to a nontrivial word in the y_i; hence the c_iy_i freely generate a free subgroup, and $y_i \mapsto c_iy_i$ is an isomorphism between two free subgroups of the base. Thus (2) is an HNN extension, and the base, hence C, embeds by Britton's lemma. The relation $c_i = ty_it^{-1}y_i^{-1}$ shows that x, y, t generate $B(C)$, and substituting this expression for c_i into the relators of C and into (2) produces a recursive presentation on x, y, t, uniformly in the presentation of C.

(2) The relators of (2) map to the relators $[t, y_i]$ of Q_+ when every c_i is sent to 1, and the map is the identity on x, y, t.

(3) The relators $[t, y_i]$ are sent to $[(1, t), (x_1^i y x_1^{-i}, 1)] = 1$, so j is a homomorphism. Let N_0 be the normal closure of $\{y, t\}$ in Q_+. Then $Q_+/N_0 = \langle x\rangle$ is infinite cyclic and the extension splits, so $Q_+ = N_0 \rtimes \langle x\rangle$. Rewriting the presentation of Q_+ over the transversal $\{x^n\}$ by the Reidemeister–Schreier method [LS, Chapter II] presents N_0 on the generators $y_n = x^nyx^{-n}$ and $t_m = x^mtx^{-m}$, $n, m \in \mathbb{Z}$, with relators the conjugates $x^m[t, y_n]x^{-m} = [t_m, y_{n+m}]$, that is, all commutators $[t_m, y_n]$. Hence $N_0 = F(y_n : n \in \mathbb{Z}) \times F(t_m : m \in \mathbb{Z})$. Now $j(y_n) = (x_1^n y x_1^{-n}, 1)$ and $j(t_m) = (1, x_2^m t x_2^{-m})$, and the elements $x_1^n y x_1^{-n}$ freely generate the normal closure of y in $F(x_1, y)$, and likewise for t; so j restricted to N_0 is an isomorphism onto the product of these two normal closures. Finally $j(x) = (x_1, x_2)$ has infinite order and $j(x)^k \notin j(N_0)$ for $k \neq 0$, because, for $h \in N_0$, the exponent sum of x_1 in $j(x^kh)$ is k. Hence j is injective on $N_0 \rtimes \langle x\rangle$. Free groups are residually finite and residual finiteness passes to direct products and subgroups, so Q_+ is residually finite. □

For $e \in \mathbb{N}$ let $Q_e = B(C_e) = F/N_e$ with C_e the group of Lemma 5.2, and write $q_e \colon F \to Q_e$. By Lemmas 5.2 and 5.3, a recursive presentation of Q_e on x, y, t is computable from e; $N_e \leq N_+$; if $e \in \mathrm{INF}$ then $Q_e = Q_+$, $N_e = N_+$ and $q_e = q_+$; and if $e \in \mathrm{FIN}$ then $H \leq C_e \leq Q_e$.

Detecting N_e inside a finitely presented group.

Lemma 5.4 (Higman embedding and the Mihailova subgroup)**.** *There is an algorithm that computes from e a finite presentation $\langle X_e \mid R_e\rangle$ of a group Λ_e, words $w_x, w_y, w_t \in F(X_e)$, and a finite generating set of a subgroup $M_e \leq F(X_e) \times F(X_e)$, such that $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$ induces an embedding $Q_e \to \Lambda_e$, and $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$. Consequently, with*

$$K_e^0 = F \times F(X_e) \times F(X_e), \qquad i_e(f) = (f, w_f, 1), \qquad L_e^0 = F \times M_e,$$

where w_f is the image of f under $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$, the map $i_e \colon F \to K_e^0$ is injective and $i_e(f) \in L_e^0$ if and only if $f \in N_e$.

Proof. Higman's embedding theorem [Hig] embeds every finitely generated recursively presented group in a finitely presented group, and its proof is effective; an explicit algorithm producing the finite presentation and the embedding words from a recursive presentation is given by Mikaelian [Mik]. Apply it to the presentation of Q_e. Let M_e be the subgroup of $F(X_e) \times F(X_e)$ generated by the pairs (z, z), $z \in X_e$, and $(r, 1)$, $r \in R_e$; then $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$ by Mihailova's lemma [Mih]. The first coordinate makes i_e

injective, and $(f, w_f, 1) \in F \times M_e$ if and only if $w_f = 1$ in Λ_e, if and only if $q_e(f) = 1$ because the embedding is injective, if and only if $f \in N_e$. $\square$

The two HNN extensions. Let $P = F(x_1, y) \times F(x_2, t)$ and $j\colon Q_+ \to P$ be as in Lemma 5.3, and put

$$K^{\mathrm{g}} = F \times P, \qquad L^{\mathrm{g}} = \{(f,\, jq_+(f)) : f \in F\},$$

and define

$$K_e = K_e^0 \times K^{\mathrm{g}}, \qquad L_e = L_e^0 \times L^{\mathrm{g}}, \qquad i\colon F \to K_e, \quad i(f) = \big(i_e(f), (f, 1)\big).$$

Let

$$\text{(3)} \quad \Gamma_e = \big\langle K_e,\ v \ \big|\ [v, \ell] = 1\ (\ell \in L_e)\big\rangle, \qquad S_e = \big\langle i(F),\ v\, i(F)\, v^{-1}\big\rangle \leq \Gamma_e.$$

Lemma 5.5 (the first HNN extension)**.**

(1) *K_e is a direct product of free groups of finite rank, hence finitely presented and residually finite; L_e is finitely generated; i is injective; and $i(F) \cap L_e = i(N_e)$.*

(2) *Γ_e is finitely presented, and a finite presentation is computable from e. The natural map from the amalgamated free product $F *_{N_e} F$, in which $n \in N_e$ in the first copy is identified with n in the second copy, onto S_e, sending the first copy to $i(F)$ and the second to $v\, i(F)\, v^{-1}$, is an isomorphism.*

(3) *There is a homomorphism $\alpha_e\colon S_e \to Q_e$ with $\alpha_e(i(f)) = q_e(f)$ and $\alpha_e(v\, i(f)\, v^{-1}) = 1$ for $f \in F$.*

Proof. (1) $K_e = F \times F(X_e) \times F(X_e) \times F \times P$ is a product of six free groups of finite rank, so it is finitely presented, and residually finite because free groups are. The subgroup L_e is generated by the basis of the first factor, the generators of M_e from Lemma 5.4, and the elements $(a, jq_+(a))$ for $a \in \{x, y, t\}$, so it is finitely generated. Injectivity of i is clear from the first coordinate. An element $i(f)$ lies in L_e if and only if $i_e(f) \in L_e^0$ and $(f, 1) \in L^{\mathrm{g}}$; the first holds if and only if $f \in N_e$ by Lemma 5.4, and the second if and only if $jq_+(f) = 1$, that is, $f \in N_+$. Since $N_e \leq N_+$, the intersection is $i(N_e)$.

(2) The presentation (3) is finite once $[v, \ell] = 1$ is imposed only for the finitely many generators ℓ of L_e, and it is computable from e because K_e, L_e and i are. It is an HNN extension of K_e with both edge groups equal to L_e and the identity as edge isomorphism. The map from $F *_{N_e} F$ to S_e is well defined because $i(n) \in L_e$ commutes with v for $n \in N_e$. A reduced word of the amalgamated free product alternates between elements of the two copies of F lying outside N_e; its image is a word $i(f_0)\, v\, i(f_1)\, v^{-1}\, i(f_2) \cdots$ in which every letter between consecutive occurrences of $v^{\pm 1}$ is some $i(f_k)$ with $f_k \notin N_e$, hence $i(f_k) \notin L_e$ by (1). Such a word contains no pinch $v^{\pm 1} \ell v^{\mp 1}$ with $\ell \in L_e$, so it is nontrivial in Γ_e by Britton's lemma.

(3) Define α_e on $F *_{N_e} F$ by q_e on the first copy and by the trivial map on the second; both kill N_e, so the definition is consistent on the amalgamated subgroup, and (2) identifies $F *_{N_e} F$ with S_e. $\square$

The second HNN extension is

$$R_e = \langle \Gamma_e \times Q_e,\ u \ \mid\ u(s,1)u^{-1} = (s, \alpha_e(s))\ (s \in S_e)\rangle, \tag{4}$$

whose edge maps $s \mapsto (s,1)$ and $s \mapsto (s, \alpha_e(s))$ are injective homomorphisms of S_e into $\Gamma_e \times Q_e$; so $\Gamma_e \times Q_e$ embeds in R_e by Britton's lemma. The presentation (4) lists the recursively enumerable relators of Q_e. Its finite replacement is

(5)
$$\widehat{R}_e = \Big\langle \Gamma_e \times F,\ u \ \Big|\ u(i(a),1)u^{-1} = (i(a),a),$$
$$u(v\,i(a)\,v^{-1},1)u^{-1} = (v\,i(a)\,v^{-1},1) \quad (a \in \{x,y,t\})\Big\rangle,$$

where $\Gamma_e \times F$ is equipped with the finite presentation of Lemma 5.5(2), the basis x, y, t of F, and the commutation relators between the two factors.

Lemma 5.6 (the finite presentation)**.** *The groups $\widehat{R}_e$ and R_e are isomorphic by the map that is the identity on Γ_e and on u and sends F onto Q_e by q_e. A code of the finite presentation (5) is computable from e.*

Proof. Both maps $f \mapsto u(i(f),1)u^{-1}$ and $f \mapsto (i(f), f)$ are homomorphisms $F \to \widehat{R}_e$, and they agree on the basis, so $u(i(f),1)u^{-1} = (i(f), f)$ for all $f \in F$; likewise $u(v\,i(f)\,v^{-1},1)u^{-1} = (v\,i(f)\,v^{-1},1)$ for all f. If $n \in N_e$, then $i(n) \in L_e$ by Lemma 5.5(1), so $v\,i(n)\,v^{-1} = i(n)$ in Γ_e, and comparing the two relations gives $(i(n), n) = (i(n), 1)$, that is, $(1, n) = 1$ in $\widehat{R}_e$. Hence the factor F of $\widehat{R}_e$ factors through $Q_e = F/N_e$, and the relations of (5) become $u(s,1)u^{-1} = (s, \alpha_e(s))$ for s in the generating set $\{i(a),\ v\,i(a)\,v^{-1} : a \in \{x,y,t\}\}$ of S_e; both sides define homomorphisms on S_e, so the relations hold for all $s \in S_e$. This defines a homomorphism $R_e \to \widehat{R}_e$; conversely, the relations of (5) hold in R_e after $F \to Q_e$, because $\alpha_e(i(a)) = q_e(a)$ and $\alpha_e(v\,i(a)\,v^{-1}) = 1$. The two homomorphisms are mutually inverse on generators. Computability of the code follows from Lemmas 5.4 and 5.5. $\square$

Lemma 5.7 (the case $e \in \mathrm{FIN}$)**.** *If $e \in \mathrm{FIN}$, then H embeds in $\widehat{R}_e$, and $\widehat{R}_e$ is not MF.*

Proof. For $e \in \mathrm{FIN}$ we have $H \le C_e \le Q_e \le \Gamma_e \times Q_e \le R_e \cong \widehat{R}_e$ by Lemmas 5.2, 5.3(1), and 5.6, and Britton's lemma for (4). Restricting MF models to a subgroup shows that subgroups of MF groups are MF, and H is not MF by [NMF, Theorem B]. $\square$

Lemma 5.8 (the case $e \in \mathrm{INF}$)**.** *If $e \in \mathrm{INF}$, then $\widehat{R}_e$ admits a tracial MF realization; in particular, it is MF.*

Proof. Let $e \in \mathrm{INF}$, so that $Q_e = Q_+$, $N_e = N_+$, $q_e = q_+$, and $\alpha_e(i(f)) = q_+(f)$, $\alpha_e(v\, i(f)\, v^{-1}) = 1$. By Lemma 5.6 it suffices to show that R_e admits a tracial MF realization, and by Lemma 4.5 applied to $\Gamma = \Gamma_e$, $Q = Q_+$, $S = S_e$ and $\tau = \alpha_e$, it suffices to produce the data of that lemma.

The group K_e is residually finite by Lemma 5.5(1), hence admits a tracial MF realization by Lemma 4.1, and Γ_e is the central HNN extension of K_e over L_e, hence admits one by Corollary 4.3.

The group $P = F(x_1, y) \times F(x_2, t)$ is residually finite; choose finite-index normal subgroups $P_1 \geq P_2 \geq \cdots$ of P with trivial intersection, let $r_n \colon P \to C_n = P/P_n$, and put $\beta_n = r_n \circ j \colon Q_+ \to C_n$ with j the embedding of Lemma 5.3(3). Since j is injective and the P_n decrease to the trivial group, every $h \neq 1$ has $\beta_n(h) \neq 1$ for all large n.

Let $G_n = (C_n \times C_n) \rtimes \langle \sigma \rangle$, where σ of order two exchanges the coordinates. Define λ_n on $K_e = K_e^0 \times (F \times P)$ by killing K_e^0 and sending (f, p) to $(r_n j q_+(f),\, r_n(p))$, and put $\lambda_n(v) = \sigma$. This respects the relations of (3): the factor L_e^0 of L_e lies in K_e^0 and is killed, and an element $(f, j q_+(f))$ of L^{g} is sent to $(r_n j q_+(f), r_n j q_+(f))$, which lies on the diagonal and commutes with σ. Hence $\lambda_n \colon \Gamma_e \to G_n$ is a homomorphism, and on the generators of S_e,

$$\lambda_n(i(f)) = (r_n j q_+(f), 1), \qquad \lambda_n(v\, i(f)\, v^{-1}) = (1, r_n j q_+(f)).$$

Let $\pi_0, \pi_1 \colon S_e \to Q_+$ be the homomorphisms that are q_+ on one copy of F in $S_e \cong F *_{N_+} F$ and trivial on the other, as in Lemma 5.5(3); thus $\pi_0 = \alpha_e$. The displayed formulas show that $\lambda_n|_{S_e} = (r_n j \pi_0,\, r_n j \pi_1)$, both sides being homomorphisms that agree on generators. Therefore

$$\ker(\lambda_n|_{S_e}) = \ker(r_n j \pi_0) \cap \ker(r_n j \pi_1) \leq \ker(\beta_n \circ \alpha_e),$$

which is the hypothesis of Lemma 4.5. □

Proof of Theorem A. By Lemma 5.6 the map $e \mapsto \widehat{R}_e$ is computable, by Lemma 5.8 the presented group is MF for $e \in \mathrm{INF}$, and by Lemma 5.7 it is not MF for $e \in \mathrm{FIN}$. Hence INF reduces to $\mathrm{MF}_{\mathrm{fp}}$ and FIN reduces to $\mathrm{NONMF}_{\mathrm{fp}}$ under computable many-one reductions. Since INF is Π^0_2-complete and FIN is Σ^0_2-complete, and since $\mathrm{MF}_{\mathrm{fp}} \in \Pi^0_2$ and $\mathrm{NONMF}_{\mathrm{fp}} \in \Sigma^0_2$ by Proposition 3.2, the two sets are complete at their levels. A Π^0_2-complete set is not Σ^0_2 and a Σ^0_2-complete set is not Π^0_2, so neither set is recursively enumerable, and in particular neither is decidable. □

Remark 5.9. The construction uses only these properties of H: it is finitely generated, recursively presented, and not MF, while subgroups of MF groups are MF. It therefore applies to any group property that is closed under subgroups, given a finitely generated recursively presented group without the property and an analogue of Lemma 4.5 for the twisted HNN extensions; the open problems below record which input is missing for soficity and for hyperlinearity.

The homomorphism V of Lemma 4.5 does not extend faithfully to R_e in the corona: the finite maps λ_n are trivial on the Mihailova factor, so W

commutes with $V(\gamma, 1)$ for every $\gamma \in K_e^0$, although u does not commute with $(\gamma, 1)$ in R_e. Injectivity is obtained in Theorem 4.2 from the universal algebra.

Open problems. We do not know whether the models of $\widehat{R}_e$ can be chosen to converge to $C^*_{\mathrm{r}}(\widehat{R}_e)$ in operator norm, so that the reduced C^*-algebra itself is MF; the argument produces only a tracial state on an abstract MF algebra. We do not know whether $\widehat{R}_e$ is hyperlinear for $e \in \mathrm{INF}$; placing hyperlinearity recognition at the same level would also require a finitely presented non-hyperlinear group, and none is known. We do not know whether an analogous family of finitely presented groups exists for soficity; such a construction would require a permutation-model replacement for Lemma 4.5. Finally, we do not know the complexity of MF recognition under the strong convergence convention of [GKM, Sch]: the norms $\|\lambda(w)\|$ in the left regular representation are not in general computable from a finite presentation, so Proposition 3.2 does not apply.

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Use of AI and formal methods

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